aluminium ‘mask’. The image that the sensor receives is immediately transferred from the image area to the opaque one – which is a storage region. So while the image-sensor is being exposed to your next great photo, the processor reads the image slowly from the storage region.

b) CMOS sensors: CMOS imagers, like CCDs, are made from silicon. CMOS, stands for Complementary Metal Oxide Semiconductor – which refers to the process in which the sensor is manufactured rather than its function. The CMOS uses complementary and symmetrical pairs of p-type (electron-accepting) and n-type (electron-losing) semiconductors made of metal oxide, for its logic functions. The CMOS sensor uses several transistors at each pixel to amplify and move the charge to the processor, using conventional wires. The signal it sends is itself, digital, and therefore does not require an Analog-to-Digital converter.

Both sensors have their pros and cons, some of which we’d like to point out below:

CCD sensors have been around a long, long time and so have reached a higher level of technical excellence and capacity than the younger CMOS.

Each pixel in a CMOS sensor has several transistors located in it. This hampers the pixels ability to collect the light because several of the photons hit the transistors and miss the photodiode completely.

CMOS sensors therefore, are more predisposed to noise. CCDs, on the other hand, produce images with very minimal evidence of visual noise.

CCDs use a lot of power on account of their process. A CMOS sensor usually consumes very little power in comparison.

CMOS sensors are cheaper in comparison to CCDs because they can be fabricated on almost any standard silicon production line.

A CMOS sensor is larger than a CCD sensor so its surface is capable of capturing more light giving higher quality pictures.